ENGAGEMENT INTELLIGENCE

Behavioral Pattern Discovery and Comparative Analytics

IT Workshop III — Project Report

Student: Ria Anne Francis
Register Number: 2025BCS0105
Department: Computer Science and Engineering

A reproducible data-analysis workflow with unsupervised clustering, model evaluation, and an interactive Streamlit dashboard.

Abstract

Engagement Intelligence studies engagement patterns in Facebook posts using the publicly available Facebook Live Sellers in Thailand dataset hosted by the UCI Machine Learning Repository. The workflow combines data-quality auditing, exploratory data analysis, time-derived features, feature-set design, log transformation, standardization, K-Means clustering, Ward-linkage Agglomerative Clustering, internal evaluation, cluster profiling, and a Streamlit dashboard. The analysis is designed to compare alternative ways of grouping posts rather than to predict a known target. The repository does not bundle the raw dataset; the notebook retrieves it from the public UCI source at runtime.

The recorded experiment compares algorithms on the same reproducible sample of 2,500 records. For the core engagement feature set at k = 2, Agglomerative Clustering achieved a silhouette score of approximately 0.5773, compared with 0.5563 for K-Means. The final full-data K-Means solution uses two clusters. These scores describe internal geometric separation only and do not prove that the clusters are objectively correct, causal, or commercially optimal.

1. Introduction

Social-media posts can generate different amounts and types of audience engagement. Reactions, comments, and shares capture related but distinct behaviors, while post format and publication time may provide additional descriptive context. This project applies data analytics and unsupervised learning to inspect such patterns in a historical dataset of posts from ten Thai fashion and cosmetics retail sellers.

The project extends exploratory analysis into a reproducible workflow: data checks are explicit, alternative feature representations are compared, model choices are evaluated quantitatively, and outputs are exported for a dashboard. The goal is descriptive pattern discovery, not causal inference.

2. Problem Statement and Objectives

The problem is to investigate whether posts can be grouped into interpretable engagement-behavior profiles and how the resulting groups change with features, preprocessing, and clustering algorithm.

• Audit dataset structure, data types, empty columns, duplicate records, missing values, and invalid engagement counts.

• Explore distributions, post-type frequencies, publication-time patterns, and relationships among engagement metrics.

• Compare a compact core-engagement feature set with a detailed reaction-mix feature set.

• Evaluate K-Means and Agglomerative Clustering using internal validation and cluster-size diagnostics.

• Interpret the final cluster profiles using raw-scale descriptive statistics.

• Provide a filterable dashboard for inspecting results and downloading selected records.

3. Dataset and Source

Dataset: Facebook Live Sellers in Thailand
Creator: Nassim Dehouche
Repository: UCI Machine Learning Repository
Dataset page: https://archive.ics.uci.edu/dataset/488/facebook%2Blive%2Bsellers%2Bin%2Bthailand
Direct CSV: https://archive.ics.uci.edu/ml/machine-learning-databases/00488/Live.csv
Research article: Dataset on usage and engagement patterns in Thailand
DOI: 10.24432/C5R60S
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The dataset contains roughly 7,050 records describing posts from ten Thai fashion and cosmetics retail sellers, collected between 2012 and 2018. Variables include post identifier, post type, publication timestamp, total reactions, comments, shares, and reaction categories such as likes, loves, wows, hahas, sads, and angrys. The notebook downloads the dataset directly from UCI; the raw CSV is intentionally not included in the project ZIP.

The dataset is historical and represents a specific set of sellers. It should not be treated as representative of all Facebook pages, industries, or current social-media behavior.

4. Tools and Technologies

• **Python:** analysis and workflow implementation.

• **Pandas and NumPy:** data cleaning, transformation, grouping, and numerical operations.

• **Matplotlib and Seaborn:** exploratory plots and cluster visualizations.

• **scikit-learn:** standardization, K-Means, Agglomerative Clustering, silhouette evaluation, and PCA.

• **Jupyter Notebook:** reproducible, stepwise analysis.

• **Streamlit:** interactive dashboard and filtered CSV export.

5. Methodology

5.1 Data acquisition and quality checks

The notebook reads the public UCI CSV URL directly. It checks dimensions, data types, missing values, unique values, duplicate rows, and empty columns. Entirely empty columns are removed; the post identifier is excluded from behavioral clustering. Publication timestamps are parsed and used to derive hour, day of week, and month for exploratory analysis. Exact duplicate rows are removed, and rows with missing or negative engagement counts are excluded from clustering rather than silently imputing values.

5.2 Exploratory data analysis

The analysis summarizes engagement counts and post-type frequencies, visualizes raw and log-transformed distributions, compares mean and median reactions by publication hour and post type, and calculates correlations among reactions, comments, and shares. Medians are reported alongside means because extreme posts can strongly affect averages. Correlations and time patterns are interpreted descriptively, not causally.

5.3 Feature engineering

Two numeric feature sets are compared. The core set contains num_reactions, num_comments, and num_shares. The detailed reaction-mix set contains comments, shares, and individual reaction categories, excluding total reactions to reduce overlap with component reaction counts. Numeric counts are transformed with log(1 + x) and standardized. A separate sensitivity analysis adds post type through one-hot encoding rather than assigning arbitrary numeric category codes.

5.4 Clustering and evaluation

K-Means is tested across k values from 2 through 8 using inertia and silhouette score. Agglomerative Clustering with Ward linkage is compared with K-Means on the same reproducible sample of up to 2,500 rows. The comparison records silhouette score, fit time, and smallest/largest cluster sizes. The final K-Means model is fit on all eligible rows using the core-engagement feature set and a fixed random seed. PCA is used only to visualize the clusters in two dimensions.

5.5 Dashboard

The Streamlit dashboard reads the exported files in outputs/. Users can filter by post type and cluster, select an engagement metric, view distributions and median profiles, compare model configurations, inspect filtered records, and download the selected records as a CSV.

6. Experimental Results

The following values are taken from the experiment outputs included in the supplied project bundle. The algorithm comparison used the same 2,500-row sample for each model/feature-set comparison.

Feature set	Algorithm	k	Silhouette	Smallest cluster	Largest cluster
Core engagement	Agglomerative (Ward)	2	0.5773	363	2,137
Core engagement	K-Means	2	0.5563	555	1,945

For the tested k = 2 core-engagement configuration, Agglomerative Clustering had a higher silhouette score than K-Means by approximately 0.0210. The scores should be compared only within the stated experiment and preprocessing. Cluster sizes are imbalanced, particularly for the Agglomerative solution, so the higher silhouette score alone is not sufficient to select a universally better model.

6.1 Final K-Means cluster profiles

Metric	Cluster 0 (n = 5,465)	Cluster 1 (n = 1,585)
Median reactions	31	275
Median comments	2	424
Median shares	0	108
Mean reactions	135.43	556.58
Mean comments	6.20	976.56
Mean shares	1.01	174.53

The final K-Means solution separates a large lower-engagement group (Cluster 0) from a smaller group with substantially higher typical reactions, comments, and shares (Cluster 1). The labels are descriptive shorthand based on medians, not intrinsic post categories. The large gap in comments and shares suggests that the second cluster contains posts with stronger discussion and sharing activity in this dataset.

6.2 Sensitivity and stability

The post-type-aware K-Means sensitivity test reached a silhouette score of 0.5421 at k = 2, lower than the recorded core-engagement K-Means score of 0.5563. This suggests that adding post type under the tested one-hot weighting did not improve the internal score in this experiment. The five recorded random-seed stability runs reported the same silhouette score (0.5537) and inertia (9028.8710). Similar scores are evidence of repeatability under these settings, but they do not guarantee that cluster labels or interpretations are stable under all data and feature changes.

7. Dashboard Functionality

• **Interactive filters:** choose one or more post types and cluster labels.

• **Overview:** selected row count, median engagement metric, post-type counts, and cluster counts.

• **Engagement analysis:** metric distribution and mean/median summaries by post type.

• **Cluster profiles:** median engagement values and post-type composition.

• **Model comparison:** compare tested algorithms and cluster counts by feature set.

• **CSV export:** download records matching the selected filters.

8. Limitations

• Unsupervised clustering has no ground-truth labels in this dataset; internal scores do not prove semantic correctness.

• Silhouette scores can favor compact geometries and should be interpreted with cluster size and profile readability.

• Engagement variables are skewed and may include extreme observations; log transformation reduces but does not eliminate their influence.

• Total reactions overlap conceptually with individual reaction categories, motivating separate feature sets.

• The dataset is historical and limited to ten Thai retail sellers, so external generalization is uncertain.

• The dashboard currently explores precomputed outputs; rerunning the notebook is necessary to regenerate results after changing the analysis or dataset.

9. Conclusion and Future Work

Engagement Intelligence provides a reproducible workflow for examining social-media engagement patterns through descriptive analytics and unsupervised clustering. It compares two feature representations and two clustering algorithms, evaluates candidate configurations, profiles the final clusters, and presents results through an interactive dashboard. In the recorded experiment, the final K-Means clusters distinguish posts with relatively low typical engagement from a smaller group with markedly higher reactions, comments, and shares. These findings describe the supplied dataset only and do not establish causes or guarantee business outcomes.

Future work could include bootstrapped stability analysis, additional clustering methods, validation on a newer and more diverse dataset, and dashboard support for rerunning selected models. Further interpretation should combine quantitative profiles with domain context and should avoid claiming that any post format causes higher engagement.

10. Reproducibility and Project Links

The notebook downloads the dataset from the official UCI direct CSV source. Install the packages listed in requirements.txt and run the notebook from top to bottom in Jupyter or Google Colab. The Streamlit dashboard can be launched with `streamlit run app.py` from the project root; the generated output files are included so the dashboard can run without rerunning the notebook.

Public dataset page: https://archive.ics.uci.edu/dataset/488/facebook%2Blive%2Bsellers%2Bin%2Bthailand
Direct CSV: https://archive.ics.uci.edu/ml/machine-learning-databases/00488/Live.csv
GitHub repository: To be inserted after the repository is published.
Live dashboard: To be inserted after deployment.

The GitHub and live-app URLs are intentionally not fabricated. Replace the two link placeholders with the actual URLs after publishing and deployment, then regenerate this report if required.

References

1. Dehouche, N. (2018). *Facebook Live Sellers in Thailand* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5R60S

2. Dehouche, N. (2020). Dataset on usage and engagement patterns for Facebook Live sellers in Thailand. *Data in Brief, 30*, 105661. https://pmc.ncbi.nlm.nih.gov/articles/PMC7218155/

3. scikit-learn documentation: K-Means, Agglomerative Clustering, silhouette coefficient, and PCA. https://scikit-learn.org/

4. Streamlit documentation. https://docs.streamlit.io/